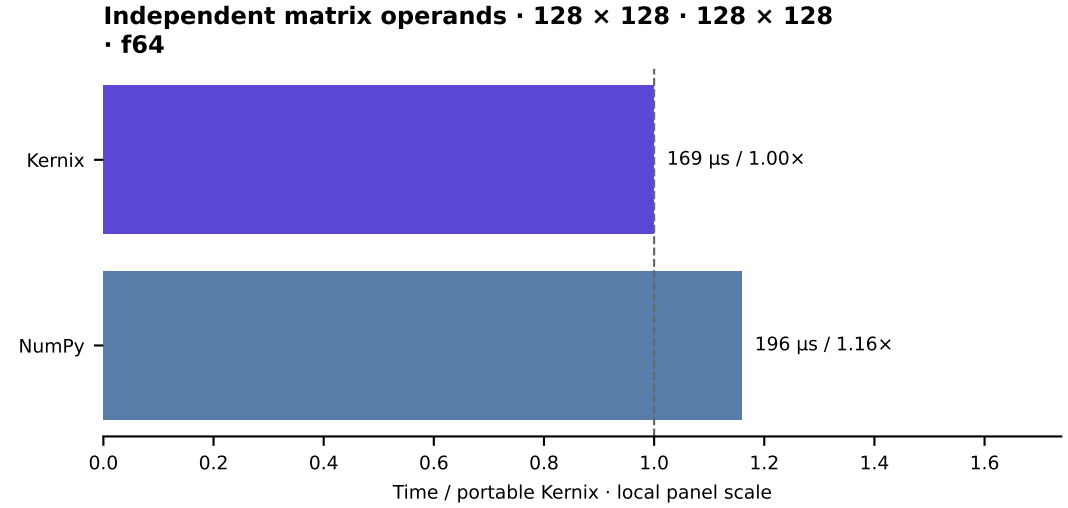
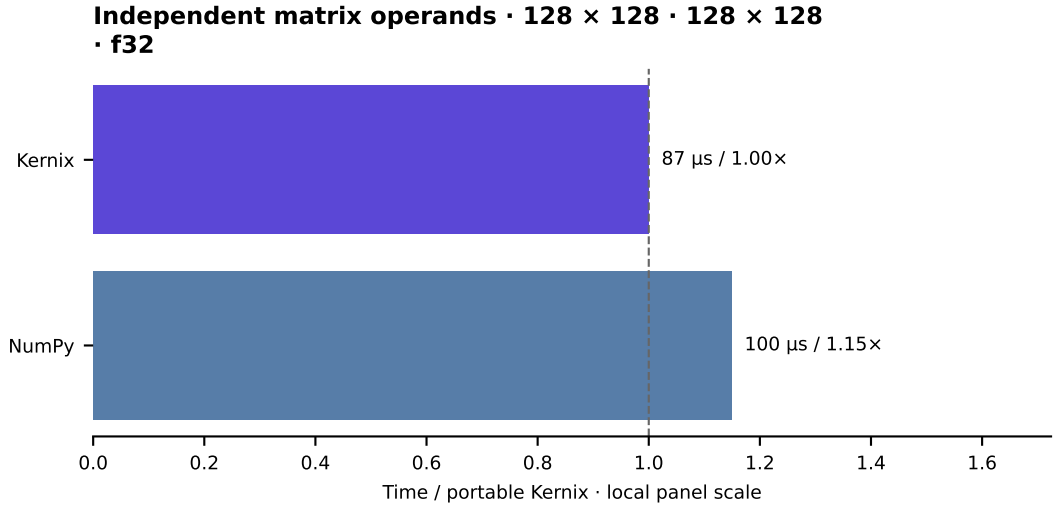
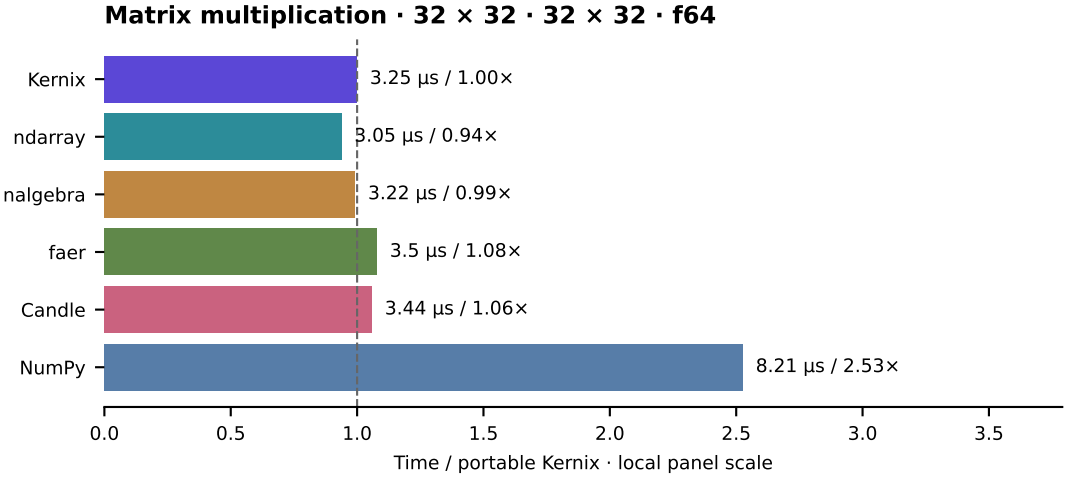
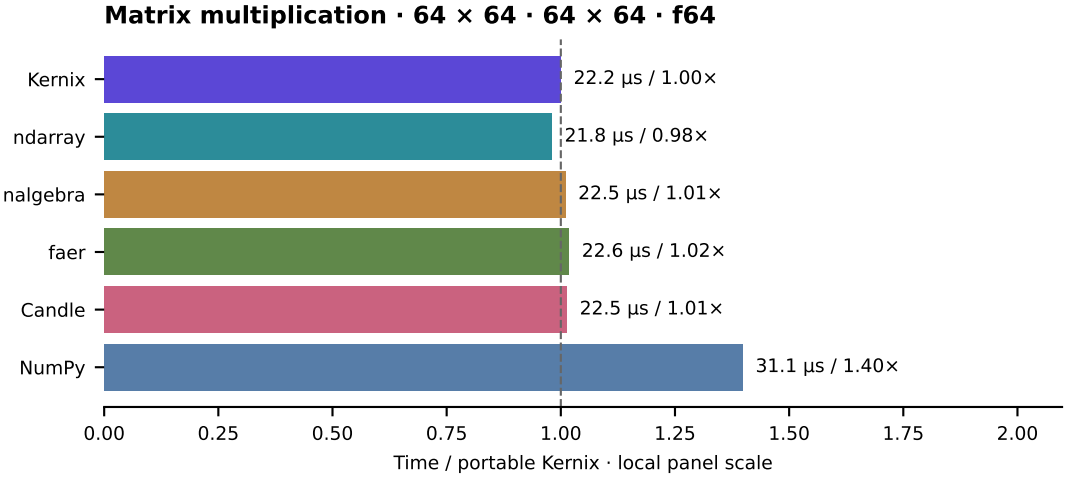
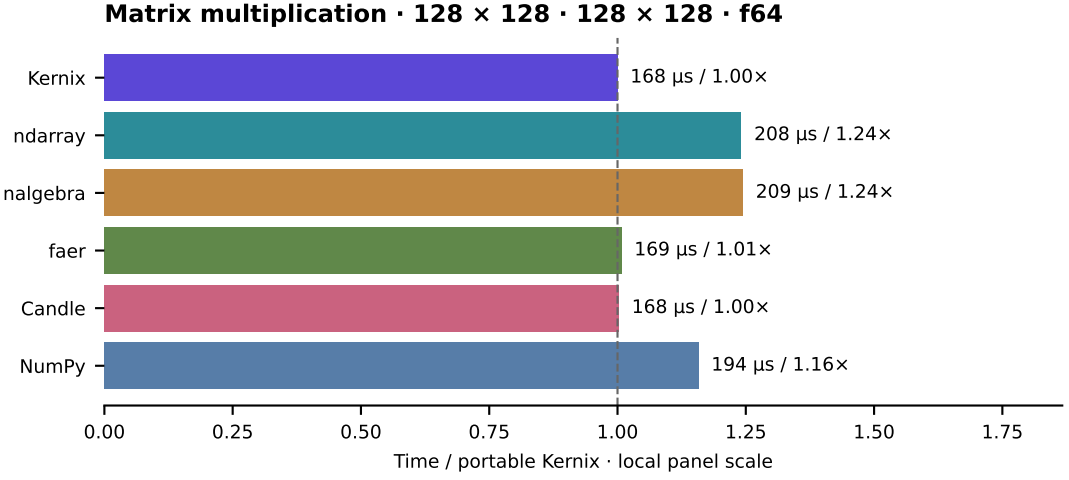
Matrix multiplication · page 1/9

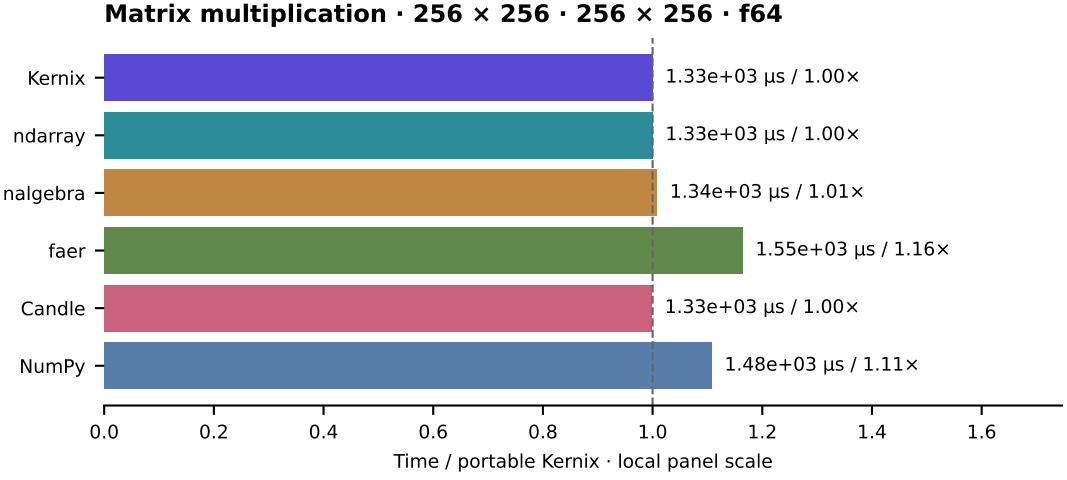




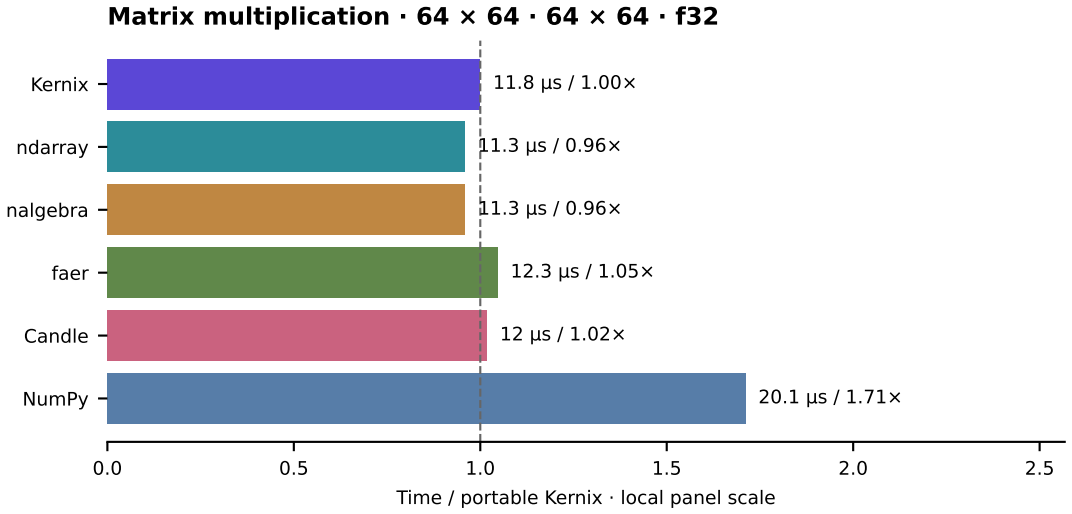


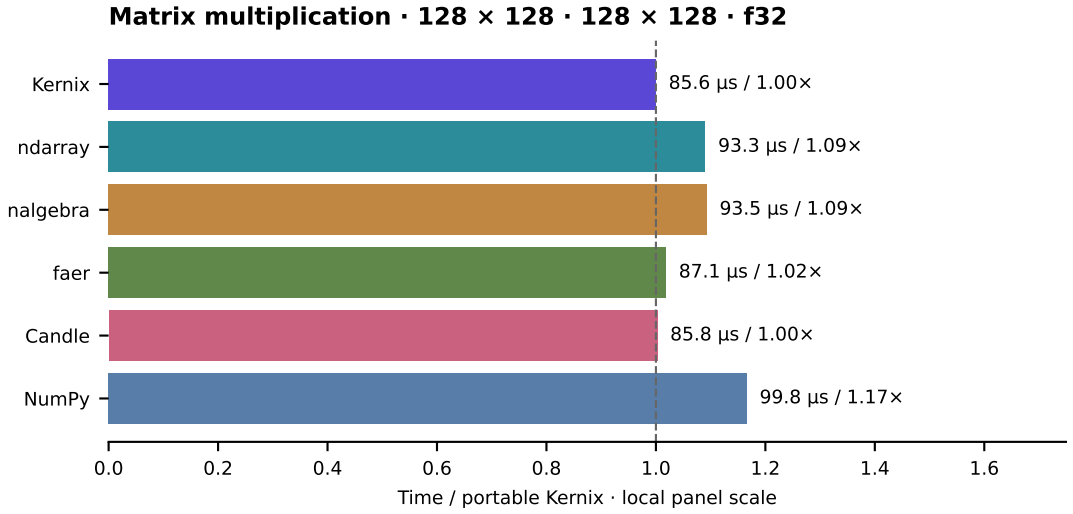


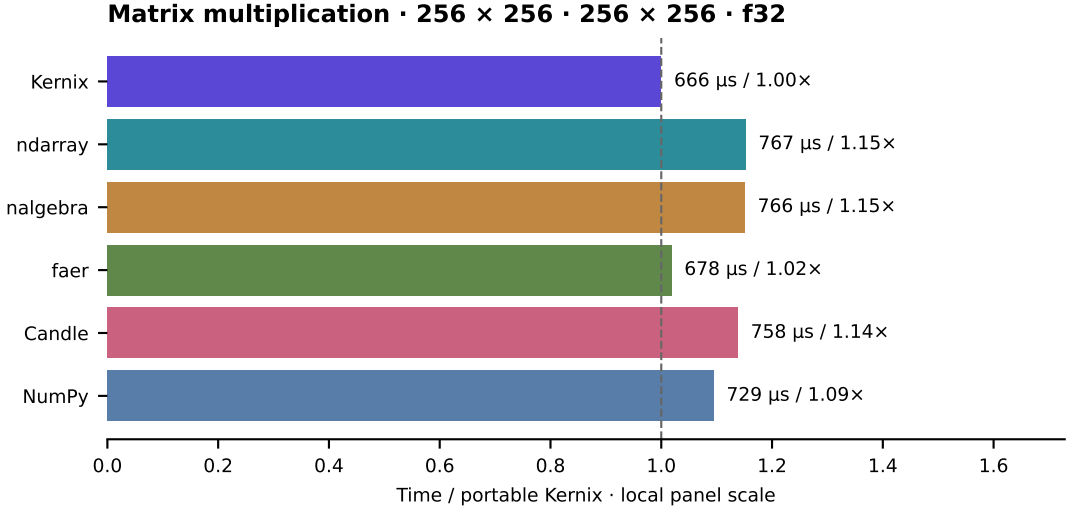


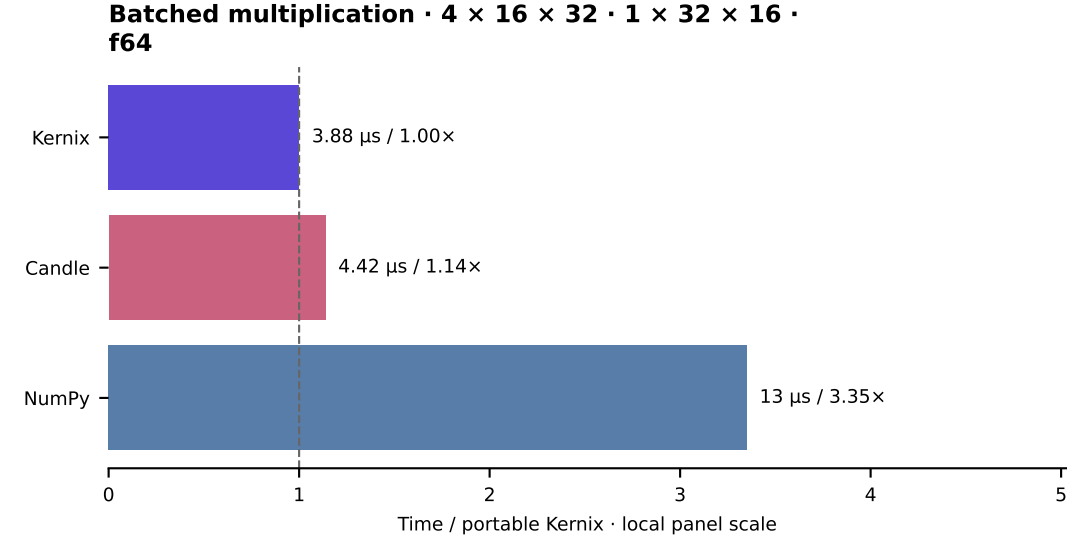


Matrix multiplication · page 2/9









Matmul · 2 workers · reused output · 64 × 64 · 64 × 64 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 256 × 256 · 256 × 256 · f64

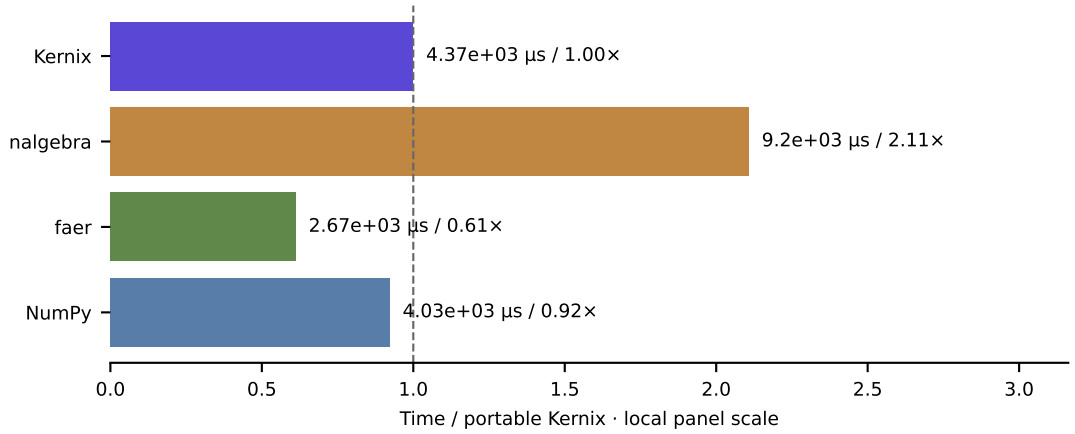
Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f32

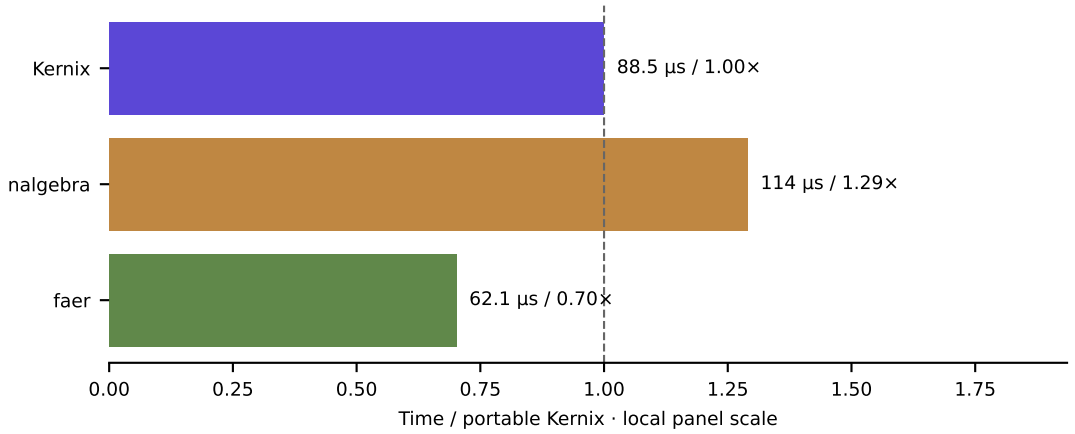
Previous-version comparison only; no peers measured

Decompositions and solvers · page 4/9

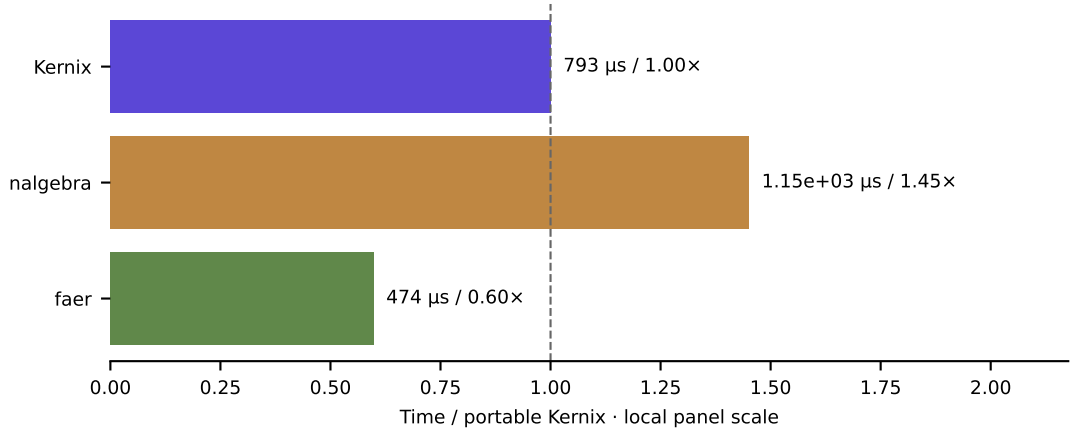
Cholesky factorization · 512 × 512 · f64



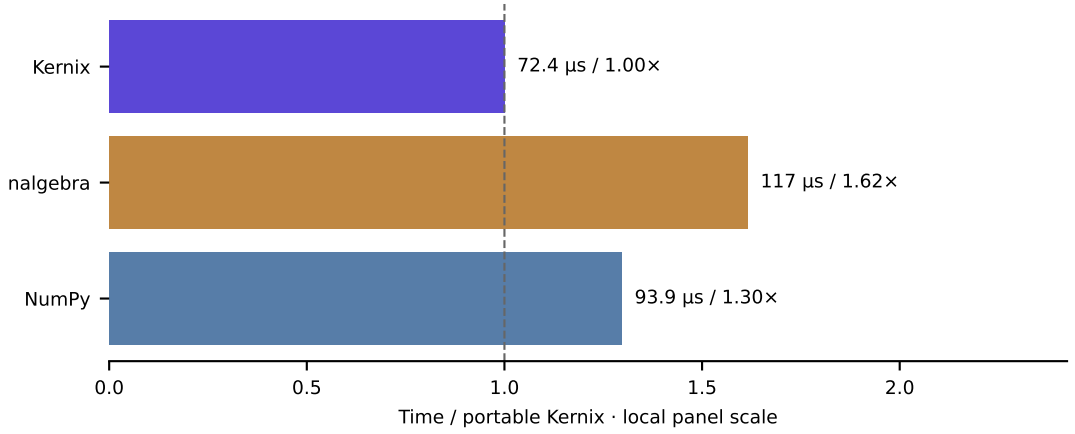
Column-pivoted QR · 96 × 32 · f64



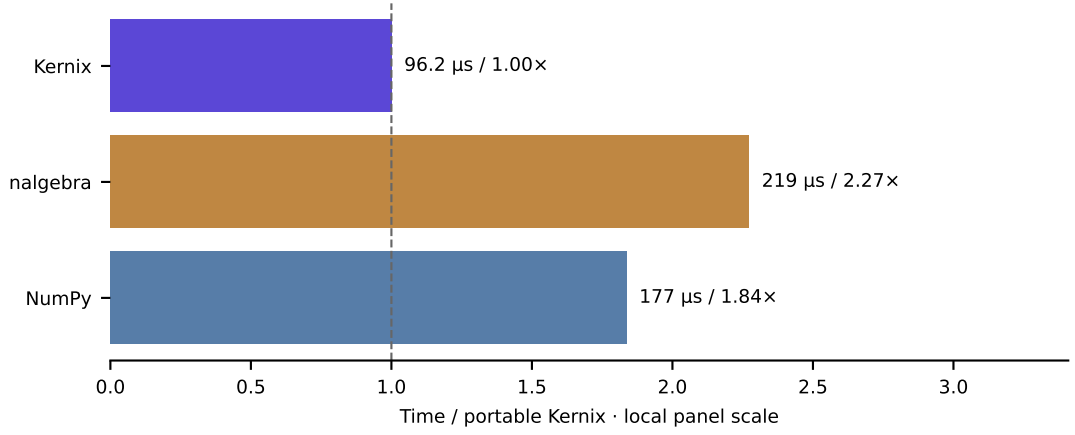
Column-pivoted QR · 256 × 64 · f64



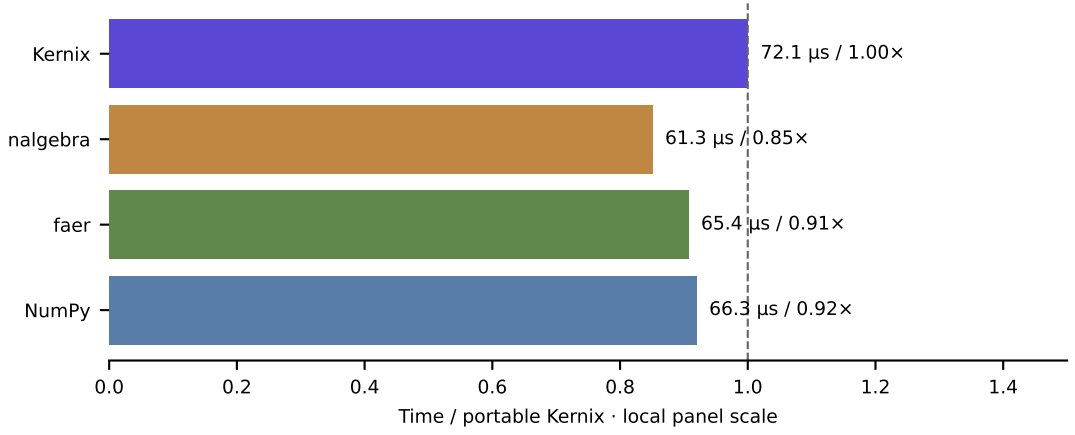
Minimum-norm least squares · 64 × 32, rank 16, 4 RHS · f64



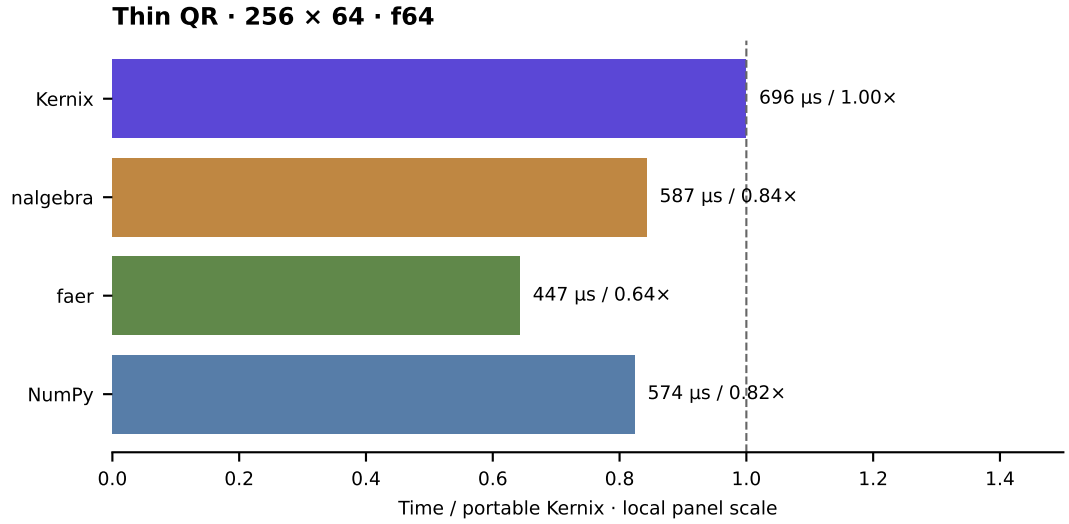
Minimum-norm least squares · 32 × 64, rank 32, 4 RHS · f64

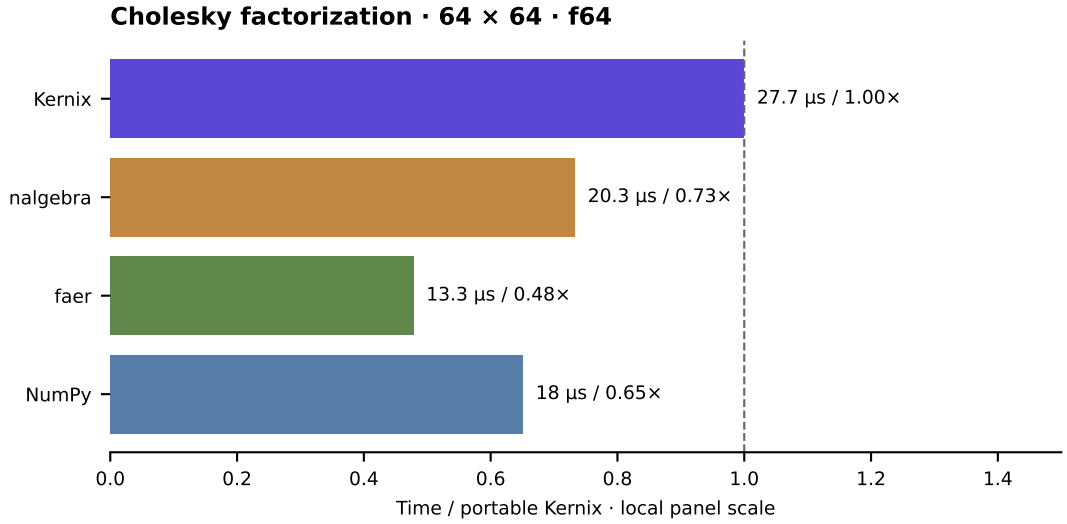


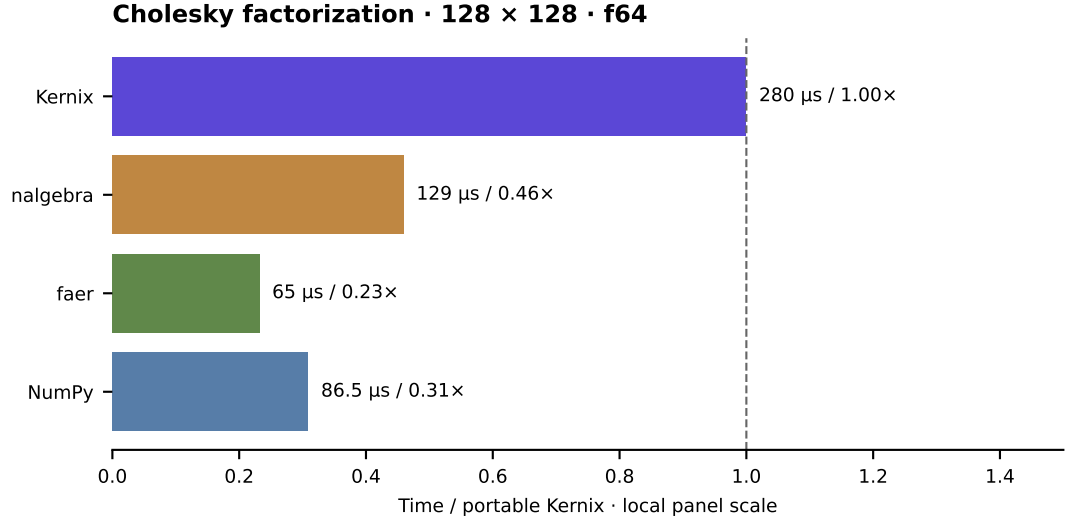
Thin QR · 96 × 32 · f64

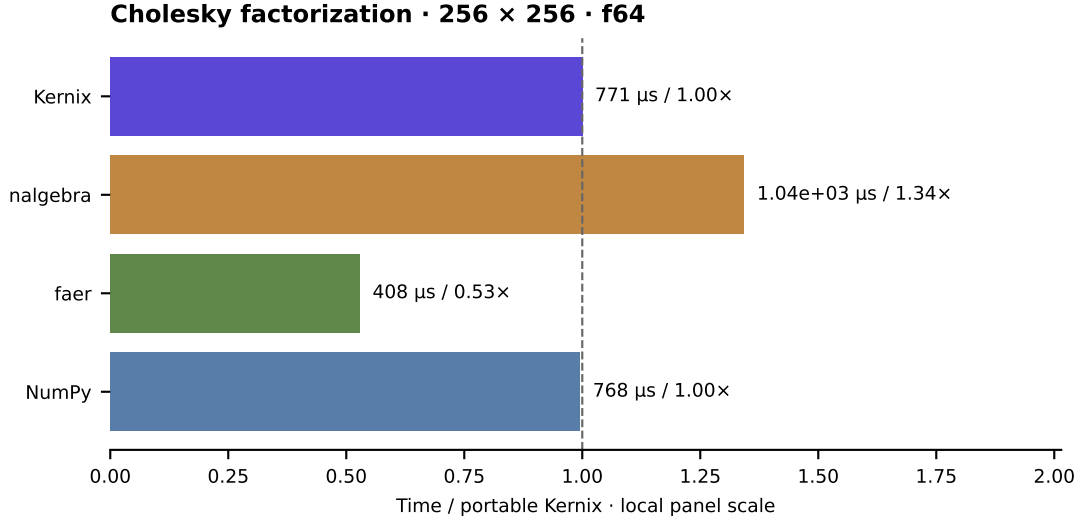


Decompositions and solvers · page 5/9

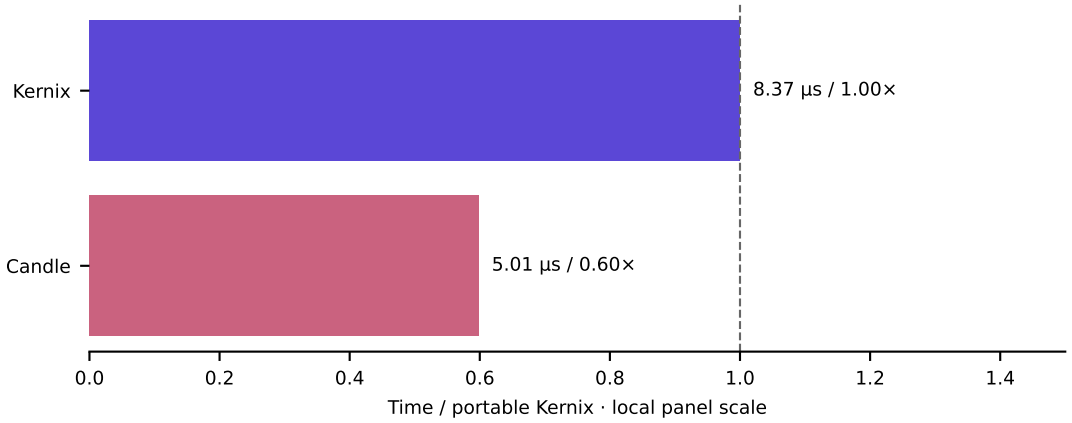




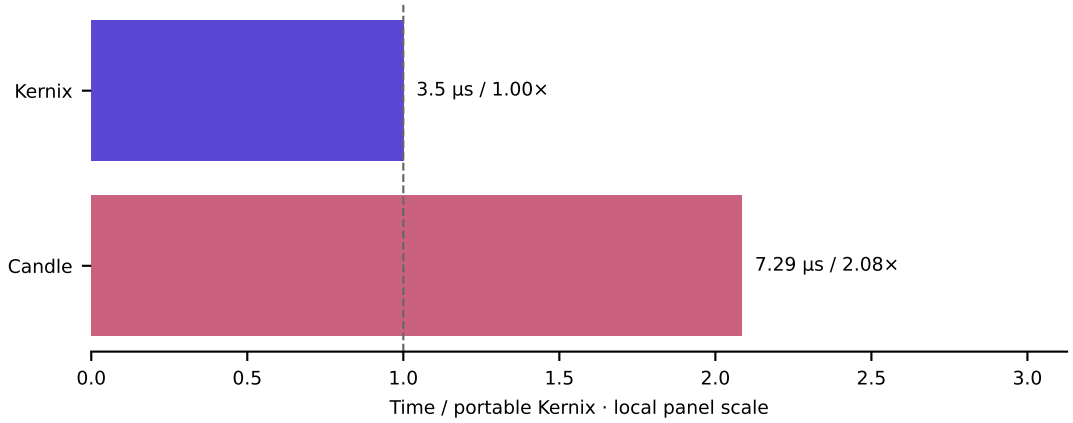




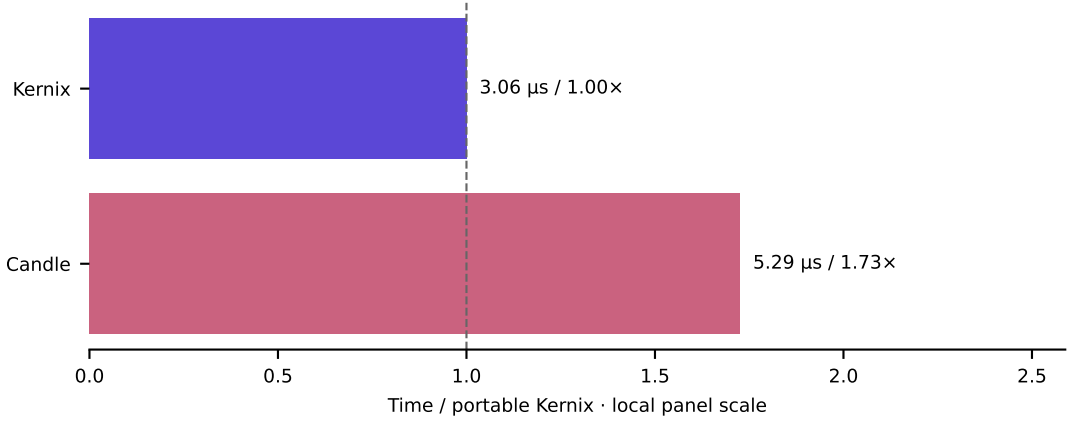
Shared-variable backward · sum(X * X), 64 × 32; prepared graph · f64



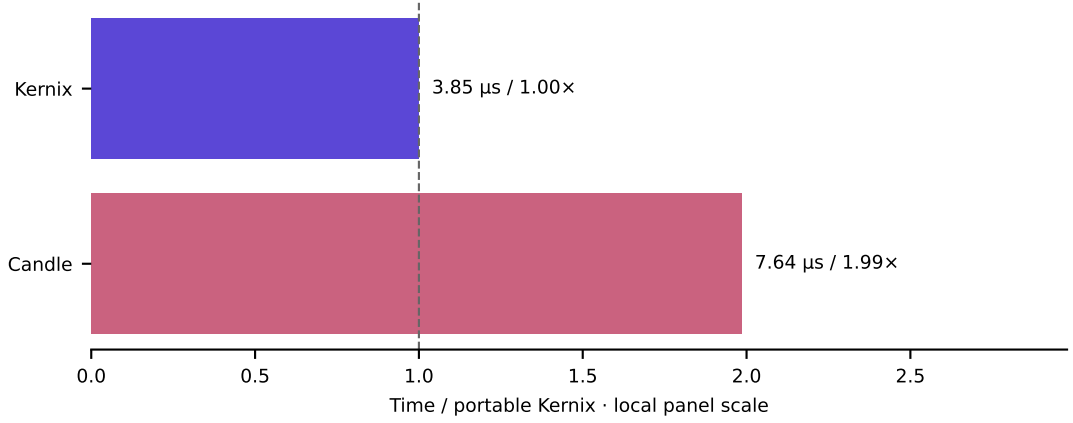
Record + backward · sum(A · W), 32 × 16 · 16 × 8 · f64



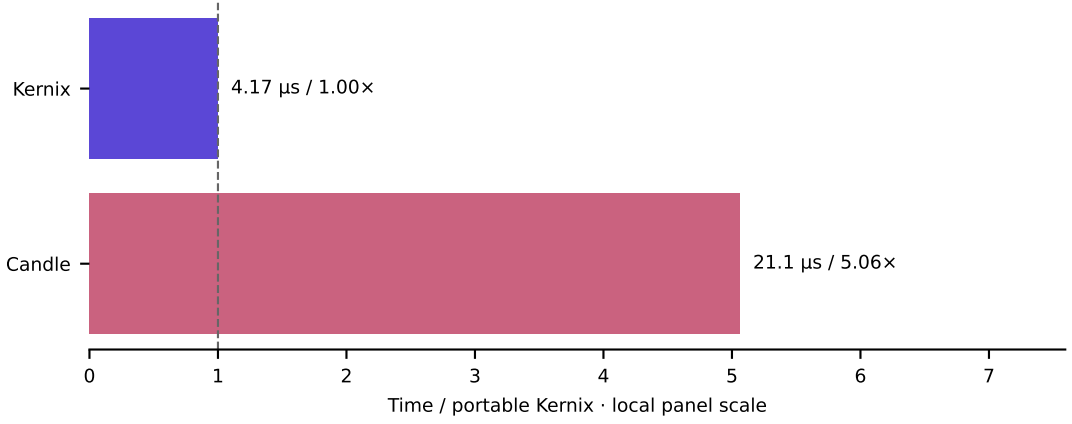
Model inference · batched normalized/sliced MLP · f64



Model recording + forward · batched normalized/sliced MLP · f64



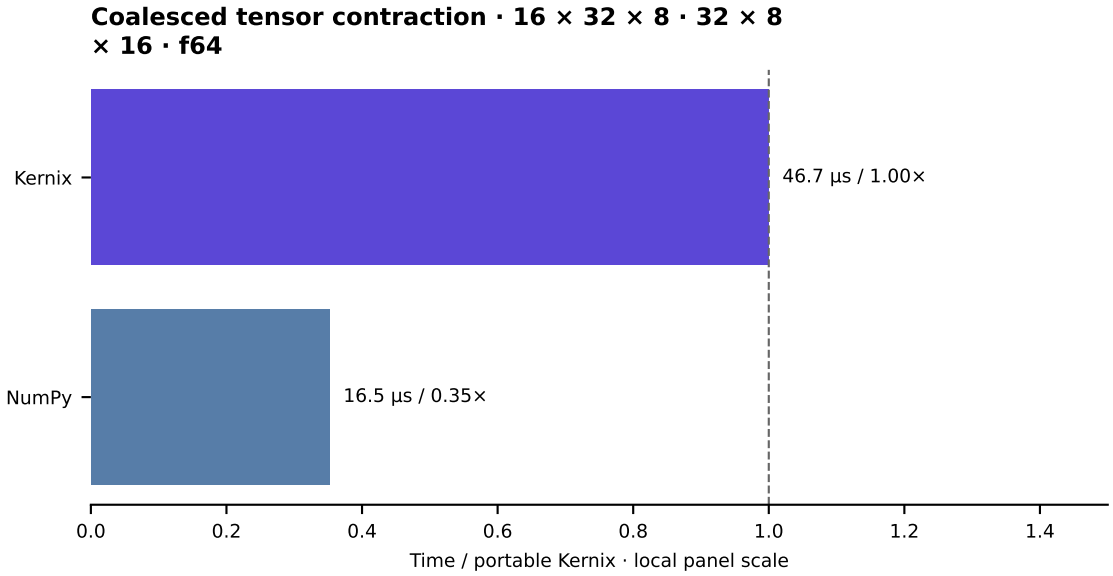
Model backward · prepared graph → fresh gradients · f64

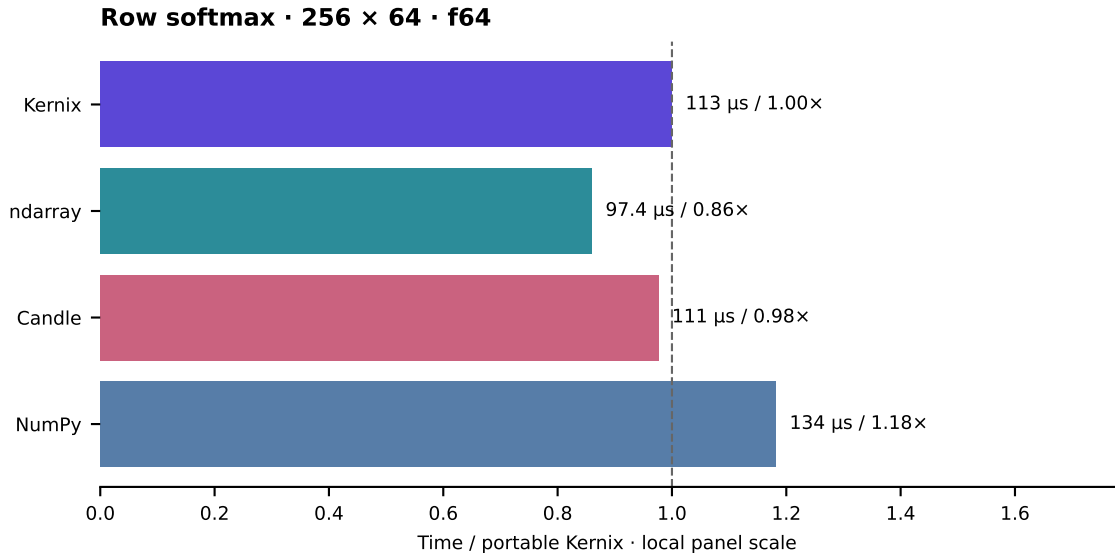


Parameter update · transactional SGD, fixed gradients · f64

Previous-version comparison only; no peers measured

Tensor contractions · page 7/9




Row softmax · 256 × 64 · f64
Kernix
ndarray
Candle
NumPy
113 µs / 1.00×
97.4 µs / 0.86×
111 µs / 0.98×
134 µs / 1.18×
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Time / portable Kernix · local panel scale

Tensor construction · page 9/9

